cannot cancel a decreasing impulse response. This means that a stable system will not have any poles with $\sigma > 0$. In other words, all of the poles in a stable system are confined to the left half of the s-plane. In fact, poles in the right half of the s-place show that the system is unstable (i.e., an impulse response that *increases* with time).

Figure (b) shows one of the special cases we have been looking for. When this waveform is multiplied by the impulse response, the resulting integral has a value of zero. This occurs because the area above the x-axis (from the delta function) is exactly equal to the area below (from the rectified sinusoid). The values for σ and ω that produce this type of cancellation are called a *zero* of the system. As shown in the s-plane diagram of Fig. 32-4, zeros are indicated by small circles ($\circ$).

Figure (c) shows the next probe we can try. Here we are using a sinusoid that exponentially *increases* with time, but at a rate *slower* than the impulse response is *decreasing* with time. This results in the product of the two waveforms also decreasing as time advances. As in (a), this makes the integral of the product some uninteresting real number. The important point being that no type of exact cancellation occurs.

Jumping out of order, look at (e), a probing waveform that increases at a *faster* rate than the impulse response decays. When multiplied, the resulting signal increases in amplitude as time advances. This means that the area under the curve becomes larger with increasing time, and the total area from $t = -\infty$ to $+\infty$ is not defined. In mathematical jargon, the *integral does not converge*. In other words, not all areas of the s-plane have a defined value. The portion of the s-plane where the integral is defined is called the **region-of-convergence**. In some mathematical techniques it is important to know what portions of the s-plane are within the region-of-convergence. However, this information is not needed for the applications in this book. Only the exact cancellations are of interest for this discussion.

In (d), the probing waveform increases at exactly the same rate that the impulse response decreases. This makes the product of the two waveforms have a constant amplitude. In other words, this is the dividing line between (c) and (e), resulting in a total area that is just *barely undefined* (if the mathematicians will forgive this loose description). In more exact terms, this point is on the borderline of the region of convergence. As mentioned, values for σ and ω that produce this type of exact cancellation are called *poles* of the system. Poles are indicated in the s-plane by crosses ($\times$).

Analysis of Electric Circuits

We have introduced the Laplace transform in graphical terms, describing what the waveforms look like and how they are manipulated. This is the most intuitive way of understanding the approach, but is very different from how it is actually used. The Laplace transform is inherently a mathematical technique; it is used by writing and manipulating *equations*. The problem is, it is easy to